

Classroom Problem-Solving Activity

Java Collections: The Library Front Desk

Course topic	Java Collections Framework
Concepts practised	ArrayList (add, remove, iterate), LinkedList used as a FIFO queue (add, remove), HashMap (put, get, containsKey)
Format	Pair work, in-class coding activity
Duration	40–45 minutes

1. Problem Statement

A small library wants a simple front-desk program with three jobs: keep track of which books are currently on the shelf, keep a waiting list of patrons who want to reserve a book that is out, and quickly look up a book's title from its ID code.

Books on shelf: [Dune, 1984, The Hobbit]

Reserve 'Dune' for: Alice

Reserve 'Dune' for: Ben

Next in line for Dune: Alice

Look up book ID 812... The Hobbit

Look up book ID 299... No book found with that ID

2. What You Need to Build

1. An ArrayList<String> shelf that starts with at least three book titles. Print it using a for-each loop.
2. A method returnBook(String title) that adds a title back onto shelf, and a method borrowBook(String title) that removes it — using ArrayList's add and remove.
3. A LinkedList<String> reservationQueue for one specific book. Add patron names to the end with add(...).
4. A method serveNextPatron() that removes and returns the patron at the front of the queue using remove(), and prints a message if the queue is empty.
5. A HashMap<String, String> catalog that maps at least four book ID codes (keys) to book titles (values) using put(...).
6. A method lookupBook(String id) that uses containsKey(...) to check the ID exists before calling get(...), printing "No book found with that ID" when it does not.
7. A main method that demonstrates all three collections: print the shelf, borrow and return a book, add two patrons to the queue and serve one, then look up one valid and one invalid book ID.

3. Try It Out

- Borrow a book that is on the shelf, then print the shelf again — confirm it is gone.
- Add two names to the reservation queue, then call serveNextPatron() twice — confirm the first name added is served first.
- Call serveNextPatron() a third time on an empty queue — confirm the program does not crash.
- Look up a book ID that is not in the catalog — confirm you see the "not found" message, not an error.

4. Discussion Questions

1. Why is an ArrayList a reasonable choice for the shelf, but a HashMap would be an odd choice for it?
2. On the LinkedList queue, remove() with no arguments removes the first item added. On the ArrayList shelf, remove() needs an index or an object. Why does the same method name behave so differently on the two collection types?
3. What would happen if you called get(id) on the HashMap without first checking containsKey(id)? Why might that be risky?
4. Could an ArrayList have been used instead of a LinkedList for the reservation queue? What would you have to be careful about if you did?